JEAN-CLAUDE JUNIOR RAYMOND

PORTFOLIO

Dual Citizenship US, CAN | Toronto, ON | jean-claude-junior.raymond@polymtl.ca | +1 (514) 242-1350 Linkedin GitHub

SUMMARY

Hardware Engineer focused on **RTL** design and functional verification for high-performance **AI** systems. Background in low-latency **FPGA** datapaths, **ASIC** flow, HW/SW Co-design, **SoC** pre-silicon verification at **AMD**, and industry-sponsored **AI hardware research** in collaboration with **Ericsson**.

EDUCATION

Polytechnique Montréal

Bachelor's Degree in Electrical Engineering | Specialization in Microelectronics and AI — August 2022 – August 2026

SKILLS

Languages: C/C++, SystemVerilog, Python, Verilog, VHDL, Bash, TCL, Assembly (RISC-V, ARM).
Verification & Infrastructure: UVM, Jenkins, Github Actions, Git, Perforce, Cocotb, Code Coverage, SVA, Constrained Random Verification, Regression, Triage
Hardware Design & Tools: Ethernet, TCP/UDP, MAC-layer, AXI (Full, Lite, Stream), SPI, UART, PCIe, Vivado, HLS, Cadence Genus/Innovus, SPICE, Matlab
AI & HW/SW Co-Design: Linux, PyTorch, CMake, HLS, Quantization, Transformer, Knowledge-Distillation, NLP, Vitis

CERTIFICATIONS

Hands-On RTL design for Networking Systems

QuickSilicon | February 2026 – June 2026

- Designed networking hardware modules in **RTL (Verilog/SystemVerilog)** for industry-inspired challenges (switches, routers, NIC datapaths)
- Implemented high-throughput packet-processing architectures involving buffering, arbitration, rate-limiting, and flow-control logic.
- Applied production **RTL** design patterns (gearboxes, wide-datapath packing, shared-buffer queues, CRC engines) used in networking **ASICs**.

WORK EXPERIENCE

Firmware Verification Engineer Intern

AMD (Advanced Micro Devices) | May 2025 – August 2026

- Authored **test plans** (~20 cases/subsystem) defining **coverage goals**, **signoff criteria**, and **verification environments**.
- Developed a **C++/Linux SoC** firmware emulation platform for the RLC subsystem, modeling microcode flows for **interrupts** & **memory** access.
- Built & deployed a CI/CD (**Jenkins**) regression farm (30+ PCs) **automating** hundreds of tests with **coverage tracking** and f**ailure root-cause**.
- Extended the **firmware emulator** with watchdog logic, **CPU** interaction modeling, **synchronization handling**, and fault-injection.
- Built **Python validation frameworks** for **automated** FW/HW **verification** across **IPs**, improving **regression coverage** and debug efficiency.
- Designed **GitHub Actions** workflows **automating regression, test execution**, and environment status; adopted by 10+ firmware teams.
- Developed an **AI agent** querying Jira, and databases to **automate failure triage** and **verification** reporting, saving hundreds of hours.

AI Hardware Acceleration Research Intern

Ericsson x Polytechnique Montréal Research Collaboration | September 2024 – May 2025

- Designed mixed-precision Transformer **accelerator** on **FPGA (Vitis HLS)** achieved **7.24 ns timing closure** with full **RTL** synthesis & **STA**.
- Optimized datapath via bottleneck analysis; reduced **DSP usage <7.3%**, achieved **7.6–8.2× energy savings**, and improved **latency/throughput**.
- **Automated Python** codegen flow converting **PyTorch** models → synthesizable **HLS/RTL** with bit-accurate NumPy **verification**.
- Integrated **accelerator** into **PCIe**-based system; enabled **2,000 inferences/sec throughput** with high-bandwidth data transfer on **Linux**.

Teaching Assistant – Programmable Logic Systems course

Polytechnique Montréal | January 2024 – May 2025

- Mentored students in the design and verification of **FPGA**-based **RTL** digital systems using **VHDL/Verilog**.
- Supervised labs on FSM design, SPI controllers, synchronous timing, and module integration; managed lab equipment.
- Guided debugging of handshake interfaces, ROM/BRAM-based architectures, and real-time audio/OLED pipelines on Artix-7 (Nexys Video).
- Reinforced best practices in synchronous **RTL** design, clock-domain considerations, and testbench-driven validation.
- Introduced students to **Linux**-based **FPGA** development workflows.

Robotics and Coding instructor

Brickz4Kids | January 2022 – May 2024

- Taught robotics, programming, and STEM concepts through hands-on engineering projects.
- Introduced students to coding fundamentals using **Python** and **C++**-style pseudocode.
- Guided students through various embedded systems projects involving sensors, motors, and basic control logic.
- Developed lesson plans to teach algorithmic thinking and problem-solving to beginners.

PROJECTS

FPGA Ultra-Low-Latency Network Packet Processor for Trading Systems

September 2025 – December 2025

- Designed ultra-**low-latency FPGA packet** processor for trading systems achieving 100–410 ns decode via speculative **parallel architecture**.
- Built **single-cycle** output datapath using speculative decoding and one-hot arbitration for deterministic **low-latency processing**.
- Developed **UVM**-style verification environment **(cocotb) with assertions, functional coverage**, and **scoreboard**-based checking.
- Implemented **constrained-random stimulus** generation and **coverage-driven verification** to exercise corner cases and protocol behavior.
- Designed **AXI-Stream datapath** for **high-throughput** packet processing with **handshake validation** and protocol compliance checks.

Pipelined RISC-V (RV32I) Processor ASIC Design

October 2024 – December 2024

- Designed 5-stage pipelined **RISC-V** (RV32I) **processor** with hazard handling, modular datapath, and control logic in **VHDL RTL**.
- Performed **static timing analysis** and **RTL** optimization to achieve **timing closure** under worst-case conditions (0.9V, 125°C).
- Completed a full **ASIC** flow (Synthesis & P&R) using Cadence Genus and Innovus on a 45nm node, achieving an area of 7,815 μm^2.
- Implemented DFT scan chains and **verified functional** correctness using assembly-based test programs and **timing-constrained simulation**.

FPGA Accelerator for AI Real-Time Handwritten Digit Recognition
January 2025 – May 2025

- Designed a fixed-point **CNN inference accelerator** in **HLS C++** for **real-time** digit recognition
- Achieved 0.318 µs inference **latency** and 95% lower power versus the **CPU** baseline, a **4,000x speedup**
- Built a **streaming HDMI** video pipeline feeding live frames into the **accelerator**, with output resolution decoupled from the input source
- Designed a triple-buffered framebuffer with independent input and output pointers to route **HDMI** frames or test patterns to the **accelerator**
- Developed MicroBlaze **C firmware** to orchestrate the **accelerator** and drive a **UART** menu for runtime **buffer routing** and resolution control

AI Accelerator Co-Design: Quantization, Distillation, and Energy Modeling
January 2026 – March 2026

- Designed integer-only CNN inference in **Python** with 2/4/8-bit quantization, achieving 2× lower MSE at 2-bit for **AI ASIC** deployment.
- Implemented QAT with STE and knowledge distillation (ViT→ResNet-32), recovering accuracy from ~10% (PTQ) to 89–92% for W2A4/W4A4
- Optimized ResNet-32 to 86.3% CIFAR-10 accuracy with 11× fewer MACs (19.6M vs. 332M) via architecture search and distillation.
- Performed energy modeling on Eyeriss-like **accelerato**r, reducing inference to 621 µJ and achieving 1.7× dataflow energy reduction.
- Applied mixed-precision quantization + Huffman coding, achieving 2.1× model compression and ≈2× lossless weight compression.

Capstone – AI-Driven EV Charging Controller
September 2025 – May 2026

- Applied **AI/ML** methods (GMM clustering, variance analysis) in **Python** to segment 500+ behavioral profiles into clusters with 99.6% purity.
- Ran quantitative **ML** experiments on accuracy, using expected-MAE to select optimal cluster configurations for **embedded AI** inference.
- Built TFT→TCN knowledge-distil. pipeline with mixed-precision quantization (2/4/8-bit), achieved 86×-603× compression for edge inference.
- Developed **simulation** framework in **Python** to validate EV charging controller, enabling hardware-in-the-loop testing on Raspberry Pi

FPGA Audio Signal Design and Processing
October 2024 – December 2024

- Designed and implemented an **FPGA-based** audio processing system for **real-time** signal generation, **modulation**, and effects processing.
- Developed VHDL modules for audio codec control, signal routing, and real-time echo and **amplitude modulation** effects.
- Integrated SPI-based communication interfaces and hardware control logic for inter-module coordination.
- Built real-time audio display and processing pipelines on FPGA, enabling **low-latency** signal manipulation and visualization.

CMOS Standard Cell Design
September 2024 – October 2024

- Designed and simulated **CMOS** logic cells (NOT gate, NOR gate, demultiplexer) using **Cadence Virtuoso**.
- Created and validated layout mask designs using **DRC** and **LVS** verification to ensure compliance with design rules.
- Performed transistor-level simulations to evaluate performance and analyze parasitic effects.
- Conducted manual placement and routing to optimize standard cell compactness in integrated circuits.
- Used **Linux** commands and Shell scripts to automate simulation workflows and manage design files.

SCHOLARSHIPS & AWARDS

University Research Scholarships for AI Hardware Acceleration	January 2026
Desjardins Bank Scholarship	October 2025
Highest University Recognition for Student Engagement	May 2025
Annual National Student Unmanned Aircraft Team Spirit Award – Lockheed Martin	May 2025
First Robotic Competition: Third place Prize	March 2019
Autonomous Award sponsored by Ford	March 2018
Betabots: Best robot design, Second Place Prize	October 2019

CLUBS & VOLUNTEERS

Co-Founder & Technical Lead
Zenith Technical Society | January 2024 – May 2025

- Co-founded technical society focused on drone and VTOL systems.
- Oversaw design planning for embedded, **FPGA**, and electrical systems, while coordinating documentation for launches and operations.
- Guided the team to 6th place nationally, 1st in Quebec, and Lockheed Martin Team Spirit Award (UAS 2025).

Embedded Systems Engineer
PolyOrbite Technical Society | September 2023 – January 2024

- Implemented SLAM (Simultaneous Localization and Mapping) algorithms to generate real-time environment maps from LiDAR data.
- Launched and configured ROS nodes for LiDAR data acquisition and system operation on **Linux**.
- Collected and recorded LiDAR sensor data in appropriate formats for mapping and analysis.

Community Outreach Volunteer
Saint-Michel Youth Forum | September 2022- January 2025

- Served meals to homeless and vulnerable community members during Christmas outreach events.

HOBBIES & INTERESTS

Sports: Basketball, Gym, American Football, Volleyball

Reading: Novels, Manga/Anime, Comics, Light Novels

Tech & Gaming: PC building, Video games, Robotics

RELEVANT COURSES

Hardware: Efficient Hardware Implementation of Deep Neural Nets |Programmable Logic Systems VLSI | Microcontrollers & Applications | Advanced Computer Architecture | Electronic Circuits

AI: Deep Learning | Machine Learning | Natural Language Processing